Supplement to "Maximum Blackouts That Keep Every Lattice Rectangle Identifiable"

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This supplement gives the computed values, evidence for computer-assisted statements, certificates, and formal checks. Theorem, lemma, and section numbers refer to the paper. Paths start from `kagey-problems/problem001` in the repository cited there.

S1 Computed values

grid	f	$a_h(m)$	by
2×2	4	1	HIB
2×3	4	12	HIB
2×4	5	32	HIB
2×5	6	80	IB
2×6	7	192	TI
2×7	8	448	TI
3×3	5	53	IB
3×4	6	224	IB
3×5	7	892	IB
3×6	8	3,420	IB
3×7	9	12,704	IB
3×8	10	45,864	IB
3×9	11	161,992	IB
3×10	12	562,912	ITD
4×4	7	316	IB
4×5	8	1,810	IB
4×6	10	26	IB
4×7	11	292	IB
4×8	12	2,118	IMD
5×5	10	8	IB
5×6	11	76	ISC
5×7	12	1,184	ISC
5×8	14	4	TSDC
5×9	15	44	TSC
5×10	16	350	TSC
5×11	17	3,004	TSC
5×12	18	24,706	TSCM
6×6	12	160	ISC
6×7	13	2,536	SC
6×8	15	8	SC
6×9	16	–	TS
6×10	17	–	TS
6×11	18	–	TS
6×12	20	–	TS
6×13	21	–	TS
6×14	22	–	TS
6×15	24	–	TS
7×7	14	7,300	SC
7×8	16	≥ 18	S
7×24	36	–	T
7×25	37	–	T
7×26	38	–	T
7×27	39	–	T

Table S1: Computed values for $h \leq m$. H: hand, T: theorem, I: ILP (value only on 3×10, 4×8, 5×6, 5×7, and 6×6), B: brute force, D: equality candidates, C: column search, S: two SAT encodings, with complete enumeration for exact counts, M: transfer counter, –: not established. On 7×8 only a lower bound for the count is given.

As a check on rectangle generation, the totals match A085582 for squares and A289832 for rectangular grids. The constructions in `jis/data/table_counts/sat_7x8_16.json` give $f(7,8) \geq 16$, completing $f(7,8) = 16$. The separate enumeration `jis/data/table_counts/sat2split_7x8_16.json` gives 18 valid constructions with 16 points. This proves only $a_7(8) \geq 18$.

S2 Evidence for the computer-assisted statements

statement	method; independent check	evidence
Cor. 32, Thm. 46(b)	candidates; brute force	`verify_narrow.py`
Thms. 25, 26, Rem. 30	SymPy; transfer matrix	`three_row_*.json`
Thm. 37, Cor. 38, Table 1	Prop. 36; candidates	`four_row_*.json`
Thms. 40(a), 45	encoding A, LRAT; encoding B	`unsat_certificates.json`, `small_cells_*`
Thms. 40(b), 43(c)	collar board; enumerators, Lean	`five_row_*family*`, `six_row_*w15*`
Thm. 40(c), Rem. 41	SAT, counter; candidate search, second counter	`five_row_*`
Thm. 43, upper bounds	structured, LRAT; second encoder, search	`unsat_certificates.json`, `six_row_sat_w*`, `six_row_excess8_*`
Thm. 43(a), lower bound	collar board; two enumerators	`six_row_collar_6x9_check.json`
Thm. 43(b), lower bound	three enumerators	`six_row_excess8_family.json`
Thm. 44	structured SAT, collar board; enumerators, brute force	`seven_row_sat_w*`, `seven_row_witness_brute.json`
Thm. 47	hand proof, Lean; full boards for $h = 5, \ldots, 9$ from the definition	`general_theorem_board_check.txt`, `GeneralTheorem.lean`
Table S1	SAT; column search	`table_counts/*`

Table S2: Computer-assisted statements, their method and check, and the evidence. Evidence files are in `jis/data`, and `verify_narrow.py` is in the problem folder.

S3 Certificates

Two independently written generators produce the same CNF clauses for the cell instances and the 70 structured instances with $m + 9$ points. We split 6×8 by its first column into 64 cases, and 7×7 and 7×8 into 128 each. The second generator checks that these splits exhaust the first-column choices. The 345 instances with $m + 8$ points on six rows, one for each labeled graph of Lemma 42 at widths 9, 10, and 11, are written by `jis/scripts/six_row_excess8_export_cnf.py`. Encoding B independently finds all 345 cases unsatisfiable. The hashes of the CNF files, the proofs, and the results of both proof checkers are in `jis/data/unsat_certificates.json` and `jis/data/six_row_excess8_certificates.json`. The proofs themselves are too large for the repository.

S4 Formal checks

Kernel-checked validity tests cover S_0, S_8, the construction in Theorem 45, the 6×15 construction in Theorem 43, and the 5×13 and 6×23 collar boards. The formal proofs also show that at most one rectangle can be hidden. The kernel also checks that $\{(1,2),(2,1),(3,2),(2,3)\}$ hides $D(1,1)$ on 4×3. Thus the restriction $m \geq 4$ in Theorem 33 is necessary.

The remaining unformalized parts are Lemma 13, Theorem 11, Lemma 42, $f(5,5) \leq 10$, Lemma 54, the seven-row constructions, and the other results in Sections 3 to 10, except Section 8, which is formalized in full (`GeneralLocal.lean` and `GeneralTheorem.lean`).